

Tailored Touch: Personalized Robot to Enhance Engagement

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Abstract. This study investigates the potential of personalized interaction strategies in human-robot interaction (HRI) by using Large Language Models (LLMs) to create an adaptive social robot. The project focuses on increasing user engagement through congruent personalization, where a robot’s interaction style aligns with an individual personality profile. Using the NAO robot as a platform, the research investigates the impact of congruent versus incongruent personalization on user engagement and satisfaction. A pilot study tested these interactions, revealing consistently high user engagement and enjoyment. While no significant differences appeared between congruent and incongruent personalization due to the limited sample size, the study highlights the potential for personalization in advancing HRI. Future research will address technical limitations and improve robot adaptability in different contexts.

Keywords: NAO · Personalization · LLM · Interaction · HRI

1 Introduction

Social robots are (semi)autonomous robots that are designed to interact with humans in socially meaningful ways, using both verbal and non-verbal communication to perform tasks and engage users. These robots have been designed for a variety of tasks and research has shown that they aid in facilitating language learning, supporting emotional development, and enhancing social interactions [11, 13, 10]. While social robots have proven effective in these areas, there remain significant opportunities for advancement. One crucial area for improvement is the personalization of human-robot interaction (HRI)[6]. By focusing on personalized interactions, we can enhance their effectiveness by providing tailor-made interactions for each individual. Personalization allows the robot to adapt its interaction style and communication approach to the unique needs, preferences, and goals of each user. Such tailored interactions can lead to higher engagement, better overall experiences, and a more meaningful connection between the user and the robot, regardless of the context in which they are used. Various methods of personalization have been proposed [16, 9], but all of them have the shortcoming that they cannot adapt to new situations, as their programs

are limited by the training data and technical implementations used by their creators. However, with the emergence of Large Language Models (LLMs), such as ChatGPT [12], a whole new adaptive research area is being explored. By incorporating detailed user personality profiles into prompts, LLMs can generate highly tailored interactions that respond dynamically to each individual’s unique personality. This personalized approach allows social robots to engage with users in a manner that is both contextually relevant and adapted to their specific traits, moving beyond the constraints of traditional personalization methods. Enhancing personalization in social robots has the potential to revolutionize fields such as education, healthcare, and customer service by providing more engaging and supportive interactions tailored to the specific personalities of individuals. Despite advancements in personalization techniques, there remains a significant research gap in developing social robots that can effectively handle novel and contextual information while interacting in a meaningful and personalized way. This project aims to address this gap by leveraging LLMs to create personalized robots that adapt their interactions based on comprehensive user personality profiles, thereby enabling more meaningful and individualized engagements across a wide range of applications. This project aims to develop such an adaptive robot by addressing the design question: "How can a robot use personalized interaction strategies to improve user engagement based on individual personality profiles?" To tackle this question, we will make use of LLMs, providing them with personality-based prompts to adjust the robot’s way of communication dynamically, enabling more engaging and personalized interactions. We will investigate whether the incorporation of congruently personalized interaction strategies in a social robot’s interactions leads to improved user engagement and overall satisfaction compared to incongruently personalized strategies. Specifically, congruent personalization is defined as aligning the robot’s interaction style with the user’s personality type (e.g. personality type 1 with interaction style 1), while incongruent personalization is defined as a mismatch between the personality type and the robot’s interaction style (e.g. personality type 1 with interaction style 3). This topic will be investigated according to the following design and research questions:

1. **Design question:** How can a robot use personalized interaction strategies to improve user engagement based on individual personality profiles?
2. **RQ1:** Does congruent personalization in a social robot lead to higher user engagement and overall satisfaction compared to incongruent personalization?
3. **RQ2:** Are there specific personality types (e.g., introversion vs. extroversion) that are more sensitive to the effects of congruent or incongruent personalization?

We formulate the following hypotheses:

1. **H1:** Congruently personalized interaction strategies will result in significantly higher user engagement ratings compared to incongruently personalized strategies, as the alignment with user preferences results in better communication and connection between the robot and the user.

2. **H2:** Users interacting with a congruently personalized robot will report higher engagement, overall satisfaction and enjoyment compared to those experiencing incongruent personalization, as the tailored interaction style reduces misunderstandings.
3. **H3:** The negative impact of incongruent personalization will be more pronounced for certain personality traits (e.g., introverts experiencing overly extroverted robot interactions) than for others.

This study contributes to understanding the nuances of HRI by focusing specifically on the impact of alignment between user personality and the robot’s interaction style. By isolating congruency as the core variable, the research will provide actionable insights into how personalization can enhance social robot design and user acceptance. These hypotheses will be tested through user studies comparing the congruently personalized and incongruently personalized robot based on user engagement scores and overall satisfaction. The remainder of this paper features the interaction- and study design portrays the results of our experiments, and discusses the implications of our findings for the future of personalized social robotics.

2 Interaction Design

2.1 Approach Overview

Our approach to designing the robot’s interaction is informed by the principles of performing arts, particularly improvisation. Improvisation emphasizes adaptability to user needs and responsiveness to contextual cues in real-time, aligning with our goal of making the robot’s interactions engaging. By using techniques inspired by improvisation, such as dynamic response generation and active listening, the robot can respond effectively to user input, creating tailored answers or follow-up questions that maintain the flow of interaction. This approach enhances the naturalness of human-robot conversation, creating a more meaningful and engaging user experience.

2.2 Design of Robot Interaction Modalities

The interaction design incorporates both input and output modalities. The robot primarily relies on auditory input modalities, using speech recognition technology to interpret user responses dynamically. This input is processed through speech-to-text conversion, enabling the robot to handle verbal input and respond accordingly accurately. For output, the robot employs both verbal and nonverbal modalities to create engaging interactions. Verbal communication is facilitated through text-to-speech (TTS) technology provided by the SIC framework, which converts dynamically generated text responses into speech. These responses are delivered in two distinct phases. During the personalization phase, responses are preprogrammed in a Dialogflow project to ensure accurate and consistent replies while gathering necessary personalization parameters from the user. In the subsequent conversational phase, responses are dynamically generated using

ChatGPT’s 3.5 model, leveraging the principles of improvisation to provide adaptive and engaging conversations. Nonverbal communication encompasses postures, gestures, and LED effects, each carefully designed to enhance user engagement. The robot uses its posture to signal the beginning and end of interactions—standing up indicates the start of the conversation, while sitting down signals its conclusion, providing clarity and context to the user [4]. Gestures are used during explanations to make interactions more expressive and engaging. These gestures are sourced from the Naoqi documentation and are continuously and randomly selected from a predefined list of twelve explanatory motions [1]. Additionally, the robot simulates human-like blinking using LED effects, which not only makes the robot appear more lifelike but also reinforces engagement by subtly indicating attentiveness while listening to the user [2]. These multimodal communication methods—postures, gestures, and LED effects—were chosen because they have been proven to enhance engagement and were achievable within the scope of the project. To ensure a seamless and interactive experience, threading was implemented to handle the different subprocesses for verbal responses, blinking, and performing gestures concurrently. This integration of verbal and nonverbal communication ensures the robot delivers a continuous and immersive interaction.

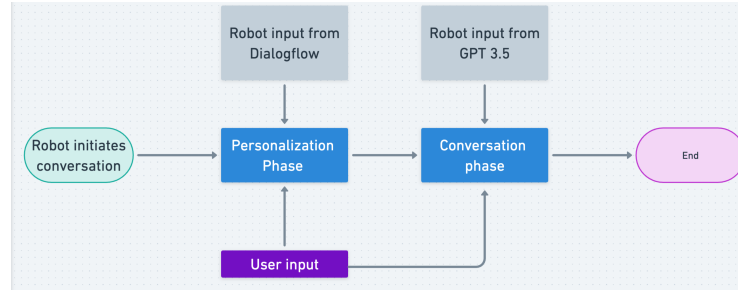


Fig. 1: Interaction Diagram

2.3 Interaction Diagram

The interaction flow is illustrated in the graphical diagram in figure 1. This flow begins with the robot initiating the interaction, after first standing up, by greeting the user. Then the personalization phase of the interaction starts in which the robot uses the input from Dialogflow and the TTS technology to react to text-converted, auditory input from the user. After the robot has gathered the necessary personalization parameters of the user (name, personality type, and topic of conversation), the robot continues into the subsequent conversational phase. In this phase, the inputs for the robot come from the user and GPT 3.5. The user input is again speech that is converted to text for better handling and the GPT input is text, which is then converted to speech by the robot. The interaction ends after this phase when the user shuts off the robot by turning the running program off. The robot will then sit down until it is turned on again.

2.4 Implementation Details

The system’s implementation integrates several software components through the SIC framework [20], which serves as the central hub connecting the robot’s hardware, software, and interaction modules. At its core, the system uses the OpenAI API to generate dynamic and contextual responses during the conversational phase, powered by ChatGPT GPT-3.5 [18]. This enables the robot to engage adaptively with users based on personalization parameters such as name, personality type, and chosen topic. The interaction with ChatGPT is facilitated by a structured prompt system. At the beginning of the interaction, a start-prompt provides ChatGPT with the required context, rules, and personalization parameters (e.g., user name, personality type and topic of conversation). During the conversational phase, repeat prompts are used each time ChatGPT is called to remind it of the rules and maintain the personalization context. This structured approach ensures consistency and relevance in the responses generated by ChatGPT. Dialogflow [14] facilitates the personalization phase by converting speech to text, classifying intents, and extracting key entities, which are then used to tailor responses and enhance user engagement. During the conversational phase, the robot continues "listening" using Dialogflow’s `GetIntentRequest` to dynamically process user input and forward it to ChatGPT via the SIC framework for response generation. Text responses are delivered through the NAO robot’s text-to-speech (TTS) module from `NaoqiTextToSpeechRequest`. Gestures and nonverbal communication are managed in two ways. In the Dialogflow phase, basic gestures are triggered by setting `animated=True` in `NaoqiTextToSpeechRequest`, syncing with verbal output. For more dynamic interactions, the SIC framework utilizes `NaoqiAnimationRequest` to execute gestures from a premade list of explanatory motions. Nonverbal cues also included simulating human-like blinking with LEDs, controlled through the `NaoLEDRequest` module of the SIC framework [19]. The last nonverbal cue was posture (e.g., standing and sitting), implemented with the `NaoPostureRequest` module. Multithreaded processing allowed the robot to simultaneously handle speech synthesis, gestures, blinking, and posture adjustments without delays, creating a smooth, interactive experience. This implementation demonstrates the integration of the SIC framework, Dialogflow, ChatGPT, and multimodal communication to create an engaging, adaptive interaction model inspired by improvisation.

2.5 Strengths and Weaknesses

The implementation effectively integrates the SIC framework, ChatGPT, and Dialogflow to enable adaptive, multimodal interaction. Strengths include dynamic response generation, seamless integration of verbal and nonverbal communication, and real-time responsiveness through multithreaded processing. The modular design of the SIC framework also facilitates scalability and future enhancements. However, the system faces challenges such as occasional response delays due to reliance on cloud-based APIs, limited synchronization between some gestures and verbal responses, and dependency on Dialogflow’s accuracy for intent recognition. Additionally, the NAO robot’s hardware constraints restrict the complexity of

actions. Future improvements could refine gesture selection and address latency to enhance overall performance.

3 Study Design

3.1 Methodology

Participants Participants will be selected from a diverse group of individuals interested in interacting with the NAO robot. Selection criteria include fluency in English, the robot’s operating language, and willingness to complete the experimental session. Participants will be categorized into personality types using the Myers-Briggs Type Indicator (MBTI) prior to the experiment. They will also complete engagement and satisfaction surveys post-interaction. For this research, the Myers-Briggs Type Indicator (MBTI) [5] was selected as the framework for constructing detailed user personality profiles due to its role in understanding individual differences. MBTI categorizes personalities into 16 distinct types, offering a structured way to capture key traits such as decision-making styles, and information-processing preferences. These traits directly influence how individuals communicate and engage with others, making MBTI an ideal choice for aligning a robot’s interaction style with user expectations

Experimental design The study employs a between-subjects design comparing two experimental conditions. First, in the Congruent Interaction Group, participants engage with the NAO robot, which adapts its interaction style to align with their MBTI personality type. The second group is the Incongruent Interaction Group where participants engage with the robot using an interaction style misaligned with their MBTI personality type. This is done by randomly picking a misaligned interaction style for a given personality type. Dependent variables are user engagement and overall satisfaction, measured through standardized questionnaires assessing enjoyability and data measured from each conversation. The Engagement Questionnaire [3] and the Overall Satisfaction and Enjoyability Questionnaire [8] are used to measure the engagement and enjoyment participants experience during their interaction with the robot. It comprises items that evaluate fun, entertainment, and pleasure, rated on a Likert scale. Higher scores on this questionnaire reflect greater overall satisfaction and engagement, indicating the robot’s perceived effectiveness in creating an engaged experience. In addition, data will be collected during the experiment which will provide objective insights into engagement. Metrics include the number and type of interactions that occur between participants and the robot, such as the frequency of participant responses and the duration of engagement. These data will serve as quantitative indicators of user involvement, complementing the subjective assessments from the questionnaires. Finally, manipulation checks will be conducted through post-experiment questions to assess participants’ perceptions of the alignment between the robot’s interaction style and their personality type, thus congruence or incongruence. This will confirm whether the manipulation effectively influenced their experience and engagement levels.

Materials and setup This experiment uses the NAO robot, which is equipped with audio input and output, a camera for visual input, and LED visual cues that provide interaction feedback to participants. The LED cues are designed to enhance engagement by signalling to participants when the robot is actively listening, providing clear and intuitive feedback about its state. Additionally, the robot’s motion capabilities, such as simple gestures or head movements, are used to make the interaction more dynamic and engaging, fostering a sense of presence and responsiveness. Several software tools are integrated to enable the robot’s functionality. Dialogflow is employed to interpret user inputs and extract personality-related parameters. The OpenAI LLM API generates verbal responses that are either aligned or misaligned with the participant’s MBTI profile, depending on the experimental condition.

The experiment takes place in a quiet and controlled environment, designed to ensure that the robot accurately processes speech and provides uninterrupted interactions. This setup minimizes disturbances and ensures the reliability of both user interactions and data collection.

Procedure Participants are first briefed about the study, including its objectives and ethical considerations, and are required to provide informed consent before proceeding. Following this, participants complete the MBTI assessment to classify their personality type, which serves as the basis for the experimental conditions. Participants are then randomly assigned to one of two groups. This will create a Congruent Interaction Group and an Incongruent Interaction Group, creating a contrast for experimental comparison.

During the interaction phase, the robot dynamically adjusts its tone, dialogue, and engagement cues, such as LED signals and motion, to reflect either congruence or incongruence based on the participant’s assigned condition. These adjustments are designed to enhance the realism and responsiveness of the interaction, providing participants with a distinctive experience. The interaction phase starts with the NAO greeting the participant and asking what their name, and personality type are. Afterwards, the participant will be asked about which topic or question they want to talk about, and the NAO will adapt and change to this specific topic.

After the interaction, participants complete two measures to evaluate their experience. The Engagement[3], Overall Satisfaction and Enjoyability Questionnaire[8] is administered to assess emotional satisfaction and hedonic engagement, capturing participants’ enjoyment and perception of the interaction. These data provide both quantitative and qualitative insights into the effectiveness of the robot’s interaction styles.

3.2 Analysis Plan

Statistical analysis will involve descriptive statistics, summarizing the scores from the Engagement Questionnaire [3] and Overall Satisfaction and Enjoyability Questionnaire [8] for both the congruent and incongruent interaction groups. This will provide an overview of the data distribution and key trends.

Engagement evaluation To measure engagement in our experiment we will use a subjective evaluation questionnaire adapted to our specific case [Fig. 4]. This first part of our questionnaire is based on the post-experiment questionnaire referenced in this paper [3]. Additionally, we will include the observers’ perceptions of engagement [17] on the Observation Sheet [Fig. 10].

Overall Satisfaction and Feelings of Enjoyment evaluation To measure the overall satisfaction we adapted our questionnaire based on the framework in this paper [8], which compares various types of interactions. We modified the questionnaire to use a 5-point Likert scale to maintain consistency throughout the entire questionnaire [Fig. 4]. Given that feelings of enjoyment influence satisfaction, we included additional questions addressing this aspect. Furthermore, we incorporated questions about whether participants felt they had sufficiently interacted with the robot, as this factor also impacts overall satisfaction.

Inferential statistics will then be applied to test the hypotheses of the study. Each variable will first be tested for normality, after which either a t-test or a Mann-Whitney U test will be used to compare the satisfaction, engagement, and enjoyability scores between the congruent and incongruent groups. The primary hypothesis is that participants in the congruent interaction group will report higher levels of engagement and satisfaction compared to those in the incongruent group. These tests will help determine whether the observed differences are statistically significant. In addition to self-reported measures, quantitative data reflecting participants’ actual engagement during the experiment will be analyzed. This includes metrics such as the number of questions asked by participants, the length of their verbal responses, and the total duration of interaction with the robot. These objective indicators provide a direct measure of how actively participants engaged with the robot, offering complementary insights into the relationship between congruence in interaction styles and user involvement. The signs of enjoyment of the participants will also be tracked by the observers during the experiments using the Human-Robot Interaction Conversational User Enjoyment Scale (HRI CUES) [17]. Moreover, sentiment analysis and linguistic complexity will be incorporated to further refine the engagement metrics [15]. Sentiment analysis can reveal emotional tones in participant responses, while linguistic complexity metrics, such as word count, parts-of-speech distribution, and sentence structure, will provide a deeper understanding of cognitive and emotional investment during the interaction. Finally, a qualitative analysis of open-ended survey responses will be conducted to identify recurring themes and insights related to participant experiences. This analysis will focus on perceptions of congruence and incongruence, providing a broader understanding of how the interaction styles influenced both user engagement and satisfaction.

4 Pilot Study

4.1 Methodology

The pilot study aims to test the feasibility and clarity of the experimental setup before conducting the full experiment. It followed the study design but was

simplified accordingly. The experiments were conducted in two controlled rooms, with the observer and the person running the code. One room was running the incongruent script and the other congruent. In total, 11 university students with 5 interacting with the NAO under the incongruent condition and 6 under the congruent condition. Participants received a printed document outlining the flow of the experiment and providing an overview of the interaction [Fig. 3]. First, they filled in the consent form [Fig. 2], and then picked the personality type that best described them. All participants engaged in a single predefined use case to ensure consistency in the collected data. They were asked to express interest in learning "How do robots learn?" The procedure was kept the same as in the study design. During the HRI, the observer noted on the observation sheet [Fig. 10] the participant's enjoyment in the interaction based on the 5-item scale which lists signs of enjoyment and dis-enjoyment[17].

- 1 Very low enjoyment — Discomfort and/or frustration
- 2 Low enjoyment — Boredom or interaction failure
- 3 Neutral enjoyment — Politely keeping up the interaction
- 4 High enjoyment — Smooth and effortless interaction
- 5 Very high enjoyment — Immersion in the conversation and/or deeper connection with the robot

When multiple cues were noticed, the observers determined the predominant level of enjoyment for each exchange. After the interaction, the participants completed a survey regarding their engagement, overall satisfaction, and enjoyment [Fig. 4]. The questionnaire primarily consisted of questions answered using a 5-point Likert scale. The pilot study evaluated feasibility (reliability and system functionality of speech recognition), clarity (of instructions and flow), and engagement.

4.2 Pilot Results

Survey Results

Exploratory Data Analysis Prior to conducting inferential statistics, Likert scale questions were mapped to correlating numerical values (Very Likely = 5, Very Unlikely = 1), and exploratory data analysis (EDA) was performed to understand the general distribution, trends, and relationships in the dataset. Descriptive statistics for each variable were calculated, including mean, median, standard deviation, and range. Boxplots and histograms were generated to visualize the distribution of responses in the congruent and incongruent groups. As seen in Table 1, descriptive statistics reveal several trends. For *Signs of Enjoyment*, the mean score was 4.00 with a standard deviation of 0.77, indicating consistently high enjoyment across participants. The average number of questions asked (*Num_ Questions*) was 2.54, with a standard deviation of 1.21, suggesting variability in participant behavior. Engagement ratings, such as *Robot was engaged during the discussion*, had a mean of 4.64 and a standard deviation of 0.50,

reflecting strong perceived engagement with minimal variation. Similarly, *Dialogue was fun* had a mean of 4.18 and a standard deviation of 0.75, highlighting an overall positive and enjoyable interaction. Participants generally rated their experiences highly, with median scores for most engagement-related variables at 4 or 5. Metrics like *Num_ Questions* (standard deviation (sd) = 1.21) and *Num_User_ Responses* (sd = 2.32) displayed greater variability, indicative of differing user interaction styles. By contrast, engagement metrics, such as *Robot understood the task* (sd = 0.54), were more consistent. Enjoyment ratings showed slightly higher variability, likely reflecting individual differences in preferences or the quality of interaction. The distributions of key metrics further support these observations. Engagement scores were predominantly high, with median values of 4 or 5 across dimensions and minimal outliers. However, a few participants rated aspects like the robot's "human-like behavior" or intelligence as low as 2. Similarly, enjoyment scores, such as *Dialogue was fun*, were positive overall, with 75% of responses scoring 4 or higher, though some responses indicated moderate levels of engagement (minimum = 2). Behavioral data, such as the number of user and robot responses, were positively skewed, with some participants contributing significantly more than others (maximum of 15 user responses). To enhance understanding, visualizations were employed. As seen in Figure 5 and Figure 6, boxplots illustrated the spread of engagement ratings, showing tight interquartile ranges (IQR) around 4–5 for most metrics, emphasizing consistency. Histograms in Figure 7 and Figure 8 revealed broader distributions for *Num_ Questions* and *Num_User_ Responses*, with peaks at the median but some extreme values. Engagement metrics occasionally exhibited bimodal distributions, clustering at 4 and 5 with dips at lower values. Density plots highlighted overlaps in user behaviors, such as the number of questions asked, while showing slight skewness toward higher engagement ratings.

Group Comparisons Engagement metrics were analyzed to assess differences between the congruent and incongruent groups. Tests for normality, performed using the Shapiro-Wilk test, determined the appropriate statistical approach. As seen in Table 2 for normally distributed variables, independent sample t-tests were conducted, while non-normal data were analyzed using Mann-Whitney U tests. Across all engagement metrics, no statistically significant differences were observed between groups. Although the results were not statistically significant, two engagement metrics approached the conventional threshold for significance. The variable *Engagement [The robot understands which topic was discussed]* yielded a p-value of 0.1028, and *Engagement [It was easy to understand the topics]* had a p-value of 0.1151. These trends suggest potential areas for further investigation in future studies with larger sample sizes.

Correlation Analysis As seen in Figure 9, correlation analysis was conducted to explore relationships between variables, with a focus on the influence of group membership (congruence) and personality type on engagement metrics. A correlation matrix revealed weak relationships between the *Congruent/Incongruent* grouping and all engagement variables, indicating minimal impact of congruence on participant perceptions or behaviors. Similarly, the categorical variable *Type*

(personality classification) showed negligible correlations with engagement outcomes. However, other patterns emerged from the exploratory correlation analysis. Higher engagement ratings were positively correlated with enjoyment metrics, indicating that participants who found the robot more engaging also reported greater enjoyment. Behavioral variables, such as the number of questions asked (*Num_ Questions*), were moderately correlated with the number of user responses (*Num_ User_ Responses*), suggesting that active questioning encouraged greater participation. Despite these trends, the limited variability in key engagement metrics likely attenuated the strength of correlations across the dataset.

Open Question Results For the question *What did you enjoy the most about the interaction with the robot?*, six of the 11 participants positively commented on the robot’s gestures, several mentioning that it added to the engagement and enjoyability. Others mentioned the topic chosen for the conversation and the fact that the robot could answer their questions informatively and teach them something new. When asked the other main open-ended question, *What aspects of the robot’s behavior could be improved?*, several participants primarily mentioned the use of face detection and the length of the robot’s speech, which five of the eleven participants found was far too long and did not allow for as many interruptions and questions as they would have liked. Lastly, of the five participants that answered the final open response question (*Any other feedback or suggestions?*), all but one spoke overwhelmingly positively about the experiment, the last reinforcing the view that the robot spoke too much during its turn.

Observation Sheet Results Experiments highlighted several repeated themes and notable observations regarding the participants’ interactions with NAO. Generally, participants displayed a mix of engagement and occasional confusion during the interactions. Most participants maintained contact with the robot and responded positively to its gestures and movements, often laughing or smiling, even when the robot made errors, such as getting their names wrong. However, repeated technical issues, including microphone sensitivity and delays in the GPT response, were problematic and led to moments of frustration or hesitancy. These issues were evident in participants with strong accents or when the robot misunderstood spoken input. A minority of participants seemed impatient or unenthusiastic to ask questions throughout the interaction and just listened passively.

5 Discussion

Statistical analysis showed no significant differences in engagement between the congruent and incongruent groups across all measured variables. This result challenges H1 and H2, which hypothesized that congruently personalized interaction strategies would lead to higher user engagement and overall satisfaction compared to incongruent strategies. The lack of statistically significant findings might be attributed to the small sample size, limiting the study’s ability to detect subtle differences. Despite this, both groups displayed consistently high levels of

engagement and enjoyment, indicating that participants generally responded positively to the robot’s interactions regardless of congruence, aligning with research emphasizing the importance of creating engaging and adaptive robot’s behavior [7]. Regarding H3, no clear evidence was found to support that certain personality types were more sensitive to congruent or incongruent personalization. Similarly, limited variability in the responses and small sample size may have influenced potential differences. Thus, further research with a larger and more diverse participant pool is necessary to explore these hypotheses. The pilot provided valuable insights into the strengths and limitations of using personalized interaction for HRI with NAO. Overall, participants showed very high levels of engagement and often responded positively to both the gestures and conversational features of NAO. However, results also pointed out various technical challenges, especially related to the microphone’s sensitivity and speech recognition features of NAO. Participants with strong accents or those positioned farther from the robot faced frequent misunderstandings, which disturbed the interaction flow. These technical issues align with prior research emphasizing the importance of robust auditory systems in social robots for effective communication [6]. Additionally, NAO’s speech duration was also an issue, with participants commenting on their limited ability to engage in a more interactive dialogue. Future work should address these technical shortcomings to enhance the robot’s functionality and user experience. For instance, incorporating advanced microphone systems and noise-cancelling technology could improve speech recognition accuracy. Enhancing the robot’s intonation and incorporating human-like pauses and breaths would make its responses more natural and conversational. The possibility of a robot being interrupted could also create a more interactive and dynamic user experience. Finally, minimizing delay and smoother transition between Dialogflow and GPT responses could significantly enhance the overall interaction quality.

6 Conclusion

This study explored the feasibility and challenges of personalized interaction using MBTI personality types in HRI using NAO robot and LLM-driven adaptations. The results showed high levels of user engagement and enjoyment, with participants positively responding to NAO’s gestures and conversational features. However, technical challenges, including microphone sensitivity, delays in GPT response generation, and NAO’s speech duration, highlighted areas for improvement. The use of MBTI as a personalization framework demonstrated potential for tailoring interactions, though the results did not reveal statistically significant differences between congruent and incongruent conditions. This could be due to a small sample size, so further analysis is recommended. Future research should focus on addressing technical limitations and incorporating nonverbal cues, such as eye contact to advance HRI and enhance engagement.

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7 Appendix

7.1 Consent Form for Pilot Study

See 2.

Consent form

This study aims to evaluate the usability, satisfaction, and overall enjoyment of interacting with a robot. During the experiment, participants will engage in a conversation with the robot, which will begin by asking for their name, personality type, and preferred discussion topic. The robot will then conduct a conversation tailored to the chosen topic. Following the interaction, participants will complete a questionnaire to assess their experience with the robot, focusing on usability, satisfaction, and enjoyment.

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Not shared

I have read and understood the information about the project, as provided in the participation information document dated _____

- I have been given the opportunity to ask questions about the project and my participation.
- I voluntarily agree to participate in the project.
- I understand I can end my participation at any time without giving reasons.
- I understand that the researcher will make observations, transcribe the conversation with the robot, collect survey responses. I understand that the data will be stored on a secured drive and will be shared with the team members of team 30 and the supervisors and professors of the course.
- I understand that the analyses and final results as outlined in the participation information document will be formatted anonymously. The researcher may present some of the findings publicly at conferences or in publications, but my identity will not be revealed in published reports.
- I understand that the data will be archived in a Github repository (https://github.com/SIR-2024/build-your-own-sir-group-30/tree/main/working_project) for 1 year for scientific integrity. I understand that other researchers will have access to this data only if they agree to preserve the confidentiality of the data.
- I understand that the data will be destroyed 1 year after the completion of the project by the researcher.
- I understand that the researcher of this project agrees that the study complies with the ethical guidelines of the Faculty of Humanities of the Vrije Universiteit (VU). I understand that if I have a complaint I should contact the Data Protection Officer of the VU (details below) and/or the Dutch Data Protection Authority (DPA) (details below).

Contact information Data Protection Officer (VU)

Vrije Universiteit Amsterdam
Data Protection Officer
De Boelelaan 1105
1081 HV AMSTERDAM
functienarisgegevensbescherming@vu.nl

Contact information Data Protection Officer (VU)

Vrije Universiteit Amsterdam
Data Protection Officer
De Boelelaan 1105
1081 HV AMSTERDAM
functienarisgegevensbescherming@vu.nl

Contact information Dutch Data Protection Authority (DPA)

Autoriteit Persoonsgegevens
PO Box 93374
2509 AJ DEN HAAG
(+31)-(0)70-8888500

I consent to participate in the research:



☐ Yes

☐ No

What is your name?

Your answer

How old are you?

Your answer

Which gender do you identify with?

☐ Female

☐ Male

☐ Non-binary/Other

☐ Prefer not to say

Fig. 2: Consent Form for Pilot Study

7.2 Script for Pilot Study

See 3.

First step:

Please fill in the consent form:



Second step:

Pick your fighter! Choose the personality type that describes you the best

Note: In case you are debating between two, just choose the one you are feeling today :)

Diplomat - The Strategic Mastermind

If you're someone who feels deeply and dreams of making the world a better place, welcome to the world of Diplomats. You have a knack for understanding people and building meaningful connections. Whether it's helping a friend through a tough time or dreaming up ways to inspire positive change, your idealism and creativity shine bright. But here's the thing: your sensitivity is both a gift and a challenge. It's okay if things don't always go as planned—your passion is what makes you truly special.

Analyst - The Organized Protector

Are you someone who thrives on strategy, logic, and solving the toughest problems? If so, you might be an Analyst! Analysts are the master planners of the world, always looking for new ways to innovate and improve. You probably love big ideas, figuring out how things work, and staying objective no matter what's happening around you. Just watch out—your focus on logic

can sometimes make you seem a little detached from others' feelings. But don't worry, your brilliance more than makes up for it!

Sentinel - The Heartful Peacemaker

Maybe you're the type of person who loves stability, tradition, and making sure things get done the right way. That means you're a Sentinel! People know they can always count on you because you're organized, reliable, and grounded. You're great at creating structure and keeping everything running smoothly. Change might not always be your favorite thing, but that's because you value consistency and take your responsibilities seriously. Your dependability is what keeps everything (and everyone) together.

Explorer - The Life of the Party

Or perhaps you're an Explorer, the adventurous spirit who's always ready to dive into the unknown! You live for excitement, new experiences, and taking life as it comes. Your quick thinking and resourcefulness mean you can handle anything thrown your way. But hey, let's be real: sometimes you get bored with routines and long-term plans. That's okay—your zest for life and ability to adapt make you someone others love to have around when the going gets tough (or when things need to get fun).

Third step:

Okay you now have chosen a personality type, the robot will start the interaction as follows:

1. NAO: *Hello there*
2. You can answer with: {hello}
3. NAO: *Hi, what is your name*
4. You answer with: {My names is ..}
5. NAO: *what is your personality type*
6. You now answer with the personality type you have picked: {My personality type is ..}
7. NAO: *Just to check your personality type is {..}*
8. You answer if the personality type is correct with a (Yes) and if its not say: (No I am ...) with the right personality type
9. NAO: *What would you like to talk about today*
10. You will answer: {I would like to learn about "How robots learn?"}
11. Now the open ended conversation will start, your endgoal is to learn about how Robots learn (e.g. are trained, take inspiration from, gather data ..)

12. After a while the conversation will end → please fill in the end survey
-

Fourth step:

Fill in the survey:



Fig. 3: Script for Pilot Study

7.3 Pilot Study Questionnaire

See 4.

Questionnaire

Part 1/4

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Not shared

* Indicates required question

Engagement *

	Strongly disagree	Disagree	Neutral	Agree	Strongly agree
The robot was engaged during the discussion of the topic.	☐	☐	☐	☐	☐
The robot understood the task.	☐	☐	☐	☐	☐
The robot could be a good partner in a cooperative discussion with a human.	☐	☐	☐	☐	☐

The robot is intelligent, understand what happens.	☐	☐	☐	☐	☐
During the discussion of the topic, the robot was showing a human-like behavior.	☐	☐	☐	☐	☐
During the discussion, the robot behaved like a child.	☐	☐	☐	☐	☐
Interaction with the robot was easy.	☐	☐	☐	☐	☐
The robot understands which topic is being discussed by the human.	☐	☐	☐	☐	☐
It was easy to understand the topic which was mentioned by the human.	☐	☐	☐	☐	☐
It was easy to see when the robot was waiting for something from the human.	☐	☐	☐	☐	☐

[Next](#)[Clear form](#)

7.4 Descriptive Statistics

Part 2/4

Overall Satisfaction

Are you satisfied with the dialogue you had? *

	Very Dissatisfied	Dissatisfied	Neutral	Satisfied	Very Satisfied
*	☐	☐	☐	☐	☐

Imagine you have been newly invited to participate in a dialogue where you talk about your own experiences and come up with new ideas. How would you feel at that moment? *

	Very Negative	Negative	Neutral	Positive	Very Positive
*	☐	☐	☐	☐	☐

Would you be likely to recommend the following initiatives to a friend or colleague? [e.g., to talk about different topics, your own experiences and think of new ideas with the robot] *

	Very Unlikely	Unlikely	Neutral	Likely	Very Likely
*	☐	☐	☐	☐	☐

*

	Strongly disagree	Disagree	Neutral	Agree	Strongly agree
Did you talk enough in the allotted dialogue time?	☐	☐	☐	☐	☐
Was the dialogue on the topic you choose convincing to you?	☐	☐	☐	☐	☐

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Next
Clear form

3/4

Feelings of Enjoyment *

	Strongly disagree	Disagree	Neutral	Agree	Strongly agree
The dialogue was fun.	☐	☐	☐	☐	☐
It was interesting to share experiences/ideas or decide on a plan.	☐	☐	☐	☐	☐

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Clear form

Open Ended Questions Part 4/4

What did you enjoy the most about the interaction with the robot?

Your answer

What aspects of the robot's behavior could be improved?

Your answer

Any other feedback or suggestions?

Your answer

Back

Submit

Clear form

Fig. 4: Questionnaire for Pilot Study

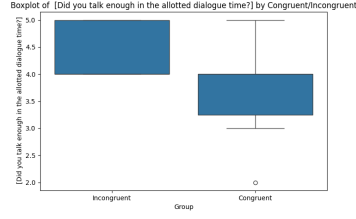
	Observation	Questions	User _ Responses	NAO _ Responses	Engaged
count	11.0	11.0	11.0	11.0	11.0
mean	4.0	2.545	10.0	10.727	4.636
std	0.775	1.214	2.324	1.272	0.505
min	3.0	1.0	7.0	9.0	4.0
25%	3.5	2.0	8.5	10.0	4.0
50%	4.0	2.0	9.0	11.0	5.0
75%	4.5	3.0	11.5	11.5	5.0
max	5.0	5.0	15.0	13.0	5.0

	Understood Task	Good Partner	Intelligent	Human-like	Child-like
count	11.0	11.0	11.0	11.0	11.0
mean	4.091	3.818	3.727	4.0	1.636
std	0.539	0.982	1.009	1.0	0.674
min	3.0	2.0	2.0	2.0	1.0
25%	4.0	3.0	3.0	3.5	1.0
50%	4.0	4.0	4.0	4.0	2.0
75%	4.0	4.5	4.5	5.0	2.0
max	5.0	5.0	5.0	5.0	3.0

	Easy Interaction	Understands Topic	Easy Topic	Easy Waiting	Fun
count	11.0	11.0	11.0	11.0	11.0
mean	3.727	3.909	3.909	3.636	4.182
std	0.786	0.539	0.944	0.924	0.751
min	3.0	3.0	3.0	2.0	3.0
25%	3.0	4.0	3.0	3.0	4.0
50%	4.0	4.0	4.0	4.0	4.0
75%	4.0	4.0	5.0	4.0	5.0
max	5.0	5.0	5.0	5.0	5.0

	Satisfied	Imagine	Recommend	Talk enough	Convincing	Interesting
count	11.0	11.0	11.0	11.0	11.0	11.0
mean	4.0	3.909	3.818	4.0	4.091	3.909
std	0.775	0.539	0.874	0.894	0.701	1.044
min	3.0	3.0	2.0	2.0	3.0	2.0
25%	3.5	4.0	3.5	4.0	4.0	3.0
50%	4.0	4.0	4.0	4.0	4.0	4.0
75%	4.5	4.0	4.0	4.5	4.5	5.0
max	5.0	5.0	5.0	5.0	5.0	5.0

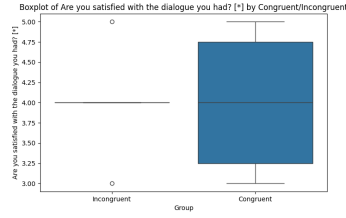
Table 1: Descriptive Statistics



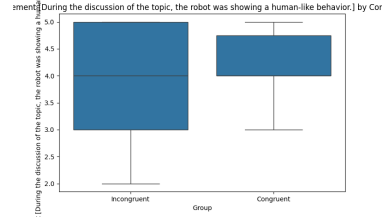
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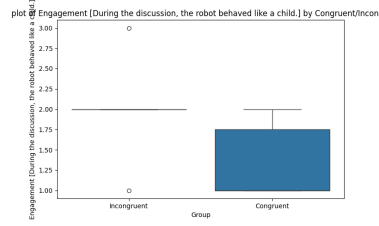
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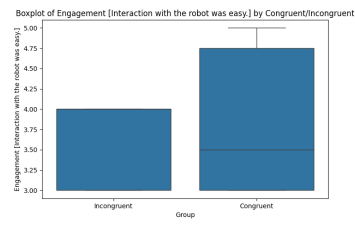
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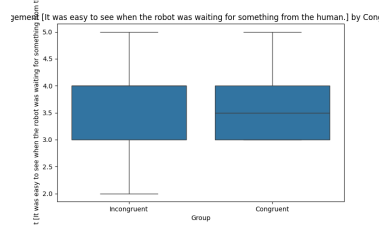
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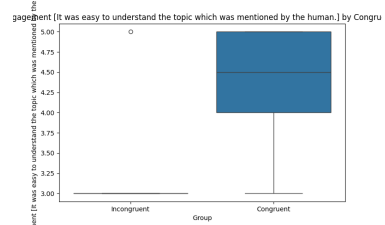
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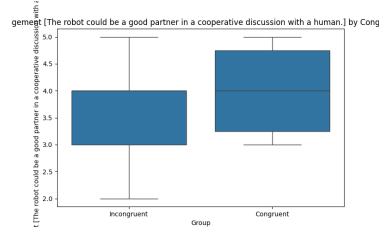
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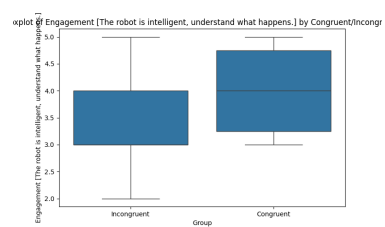
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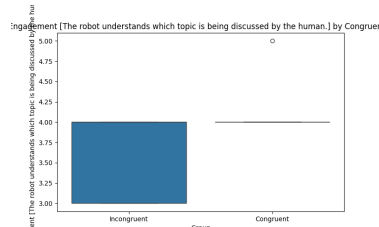
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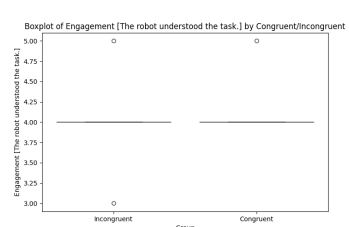
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(j) 5b

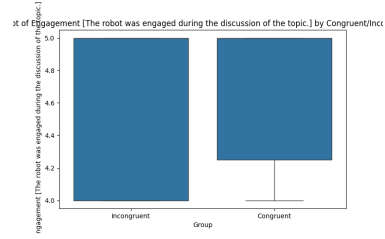


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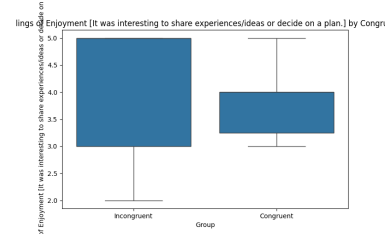


(l) 6b

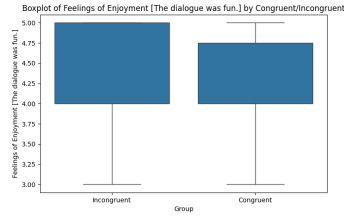
Fig. 5: Boxplots pt.1



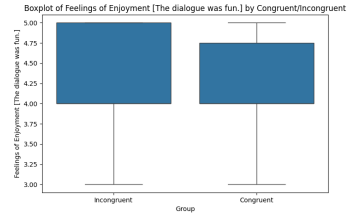
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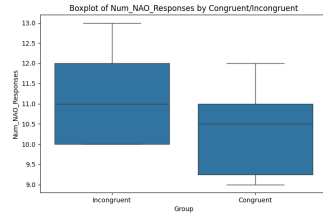
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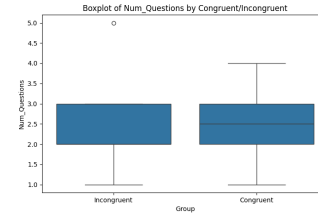
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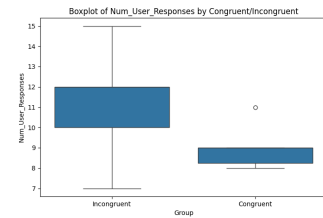
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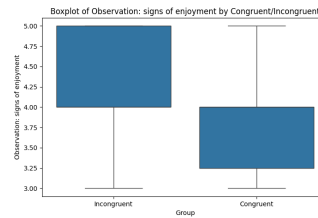
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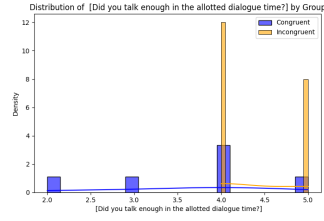


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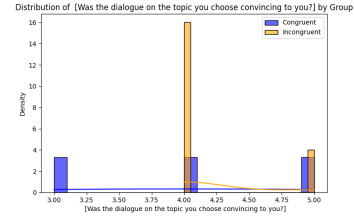


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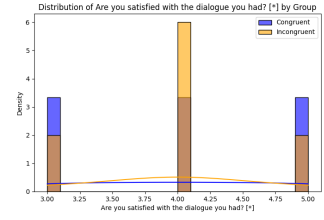
Fig. 6: Boxplots pt.2



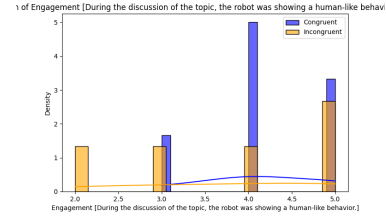
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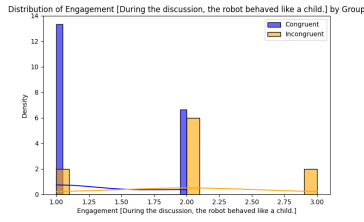
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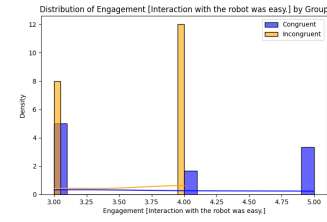
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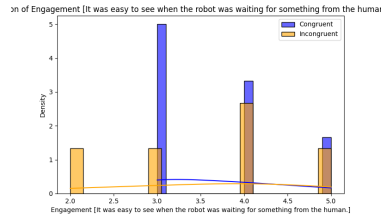
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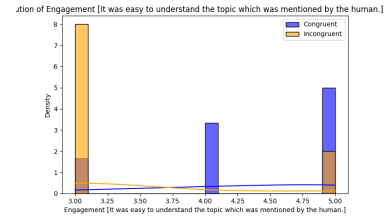
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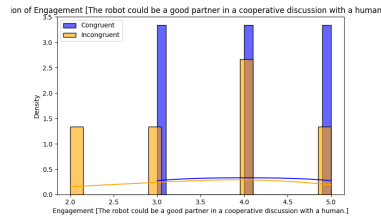
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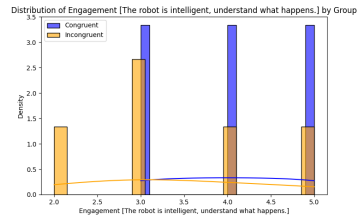
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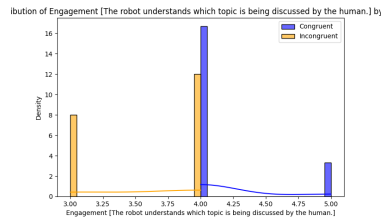
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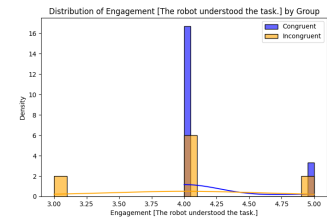
(i) 5a



(j) 5b

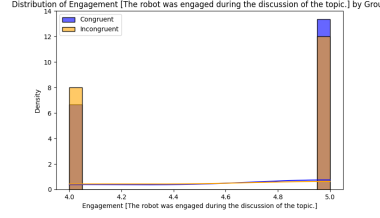


(k) 6a

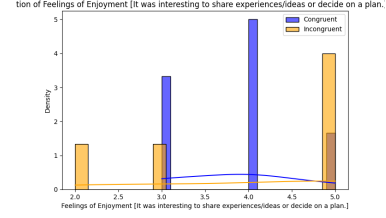


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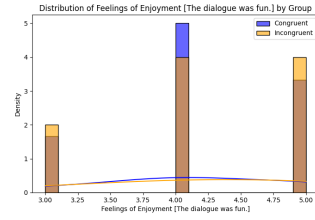
Fig. 7: Histograms pt.1



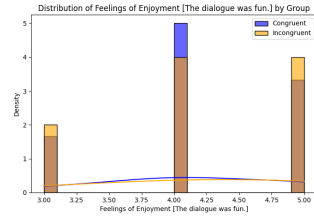
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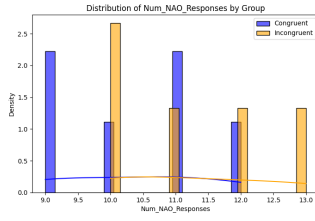
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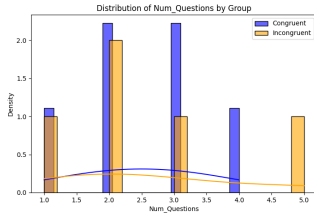
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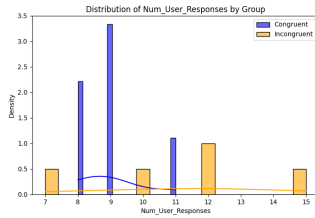
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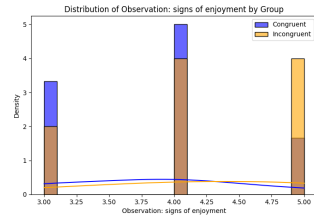
(e) 3a



(f) 3b



(g) 4a



(h) 4b

Fig. 8: Histograms pt.2

7.5 Statistical Analysis

Variable	Test Used	Test Statistic	P-Value
Engagement [The robot understands which topic ...]	Mann-Whitney U	22.500	0.103
Engagement [It was easy to understand the topic...]	Mann-Whitney U	23.500	0.115
Engagement [During the discussion, the robot b...]	Mann-Whitney U	7.000	0.130
Num_User_Responses	T-Test	-1.579	0.176
[Did you talk enough in the allotted dialogue...]	Mann-Whitney U	8.500	0.227
Num_NAO_Responses	T-Test	-1.133	0.288
Engagement [The robot is intelligent, understand...]	T-Test	0.957	0.368
Observation: signs of enjoyment	T-Test	-0.757	0.470
Engagement [The robot could be a good partner ...]	T-Test	0.638	0.542
Engagement [During the discussion of the topic...]	T-Test	0.556	0.598
Feelings of Enjoyment [It was interesting to s...]	Mann-Whitney U	12.500	0.703
Imagine you have been newly invited to partic...]	Mann-Whitney U	13.000	0.727
Engagement [The robot understood the task.]	Mann-Whitney U	17.000	0.727
[Was the dialogue on the topic you choose con...]	Mann-Whitney U	13.000	0.762
Engagement [Interaction with the robot was easy.]	Mann-Whitney U	16.500	0.844
Num_Questions	T-Test	-0.125	0.904
Engagement [The robot was engaged during the d...]	Mann-Whitney U	16.000	0.913
Engagement [It was easy to see when the robot ...]	T-Test	0.109	0.916
Would you be likely to recommend the followin...]	Mann-Whitney U	14.000	0.920
Feelings of Enjoyment [The dialogue was fun.]	T-Test	-0.069	0.947
Are you satisfied with the dialogue you had? [...]	T-Test	0.000	1.000

Table 2: Results of Statistical Analysis

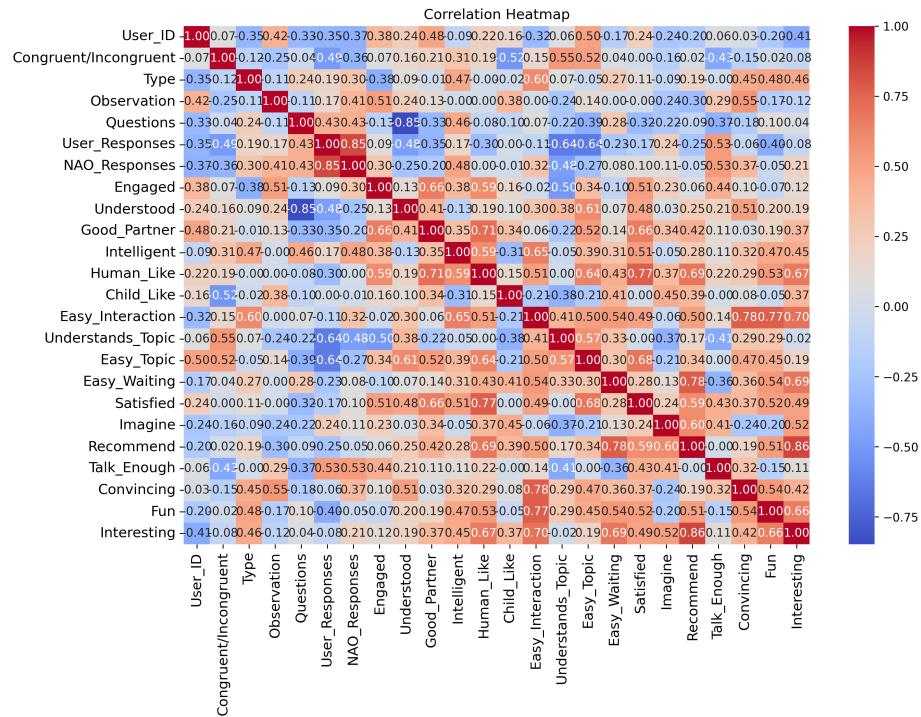


Fig. 9: Correlation Matrix

7.6 Observation sheet

Observation Sheet	
Part 1: Participant Information	
Participant's name & ID	
Duration of interaction (in minutes)	
Group	☐ Congruent (correct personality type) ☐ Incongruent (incorrect personality type)
Part 2: Signs of Enjoyment	
Rate the User's Enjoyment Level (Cross one):	☐ 1 (Very Low) 😞 ☐ 2 (Low) 😐 ☐ 3 (Neutral) 😐 ☐ 4 (High) 😊 ☐ 5 (Very High) 😄
Part 3: Notes	
Describe Key Moments in the Interaction: (E.g., specific emotional responses, changes in behavior, key dialogue moments) Additional Observations (Optional): (E.g., robot behavior affecting the interaction, unexpected events, participant feedback) Relevant Context (Optional): (E.g., environmental factors, participant's initial mood)	-

Fig. 10: Observation sheet used during experiments for each participant

7.7 Individual Contribution

Floortje Throughout the course, I contributed to various aspects of our project, focusing on technical implementation, and the design document. During the first weeks, I worked on setting up the SIC framework and debugging issues related to GitHub and the framework's compatibility with different environments. I collaborated with team members and the TA to resolve these problems, ensuring the framework was functional on my system. I set up a working OpenAI key for integrating GPT with the NAO robot and tested its interaction capabilities. This involved creating a hardcoded version combining GPT, Dialogflow, and NAO components. I also focused on enabling the system to save conversations in JSON format for further analysis, which was one of our features for capturing experimental data. In addition to technical tasks, I contributed extensively to the first setup of the design document. I made a setup for the Interaction Design, Study Design, and Pilot Study sections. After this, I finalized the Study Design Section, wrote the Abstract, and helped implement the Appendix. I conducted background research and explored text analysis techniques to enhance the quality of our experiment's evaluation. Furthermore, I helped with filling in the ethical self-check. Finally, I took part in running the pilot study and the main experiment, observing participants and analyzing results. My work included preparing materials for our experiment. After the experiments, I made a quick separate CSV file with the overview of all results anonymized, according to the ethical self-check. Finally, I helped finalize the design document and assisted with the preparation of the presentation and poster.

Aelys During the course, my primary focus was on the technical implementation, though I also contributed to the design document and actively participated in group discussions and meetings. In the first two weeks, I dedicated time to developing the technical skills and knowledge required to work with the framework and robot. Although I encountered several challenges with the framework and faced various bugs, I successfully resolved them with the assistance of the TA and my groupmates. Additionally, I took responsibility for creating and organizing the design document, which included author details, institutional information, section guidelines, and rubric criteria to ensure clarity and completeness. As the project progressed, I primarily worked on the GPT implementation. Initially, this involved working on the desktop setup, and later transitioning to integrating GPT with the robot. Once the GPT component was functional, I collaborated closely with Nika on code integration and brainstorming hardcoded Dialogflow solutions. This phase required multiple rounds of troubleshooting. To mitigate risks, I also developed a backup code solution as a contingency plan when the original implementation faced significant issues. In the final week, during the pilot study, I worked alongside Floortje and Nika to run and debug the code during participant sessions for the congruent group, ensuring the sessions ran smoothly. After the pilot study was complete, I analyzed the questionnaire responses and wrote the results section, including relevant tables and figures in the appendix due to their size. Finally, I helped finalize the design document, ensuring that tables, figures, and headings were properly formatted.

Nika Throughout the course, I contributed to both the technical implementation and the design document. Early in the project, I worked on installing the SIC framework and completing tutorials 1, 2, and 3 to familiarize myself with the system. Although I faced initial challenges with starting Redis and navigating the GitHub setup, I overcame these issues with the help of the TA and my groupmates. During this period, I also participated in brainstorming to define our interaction problem, focusing on how the robot would communicate, key metrics for personalization, and how could we measure engagement. With the help of Sebastian, we have decided to choose the MBTI, and suggested it to the group. My main task was creating an agent in Dialogflow and later hard-coding it. This is where things got tricky as I was struggling for two weeks trying to save entities as variables to be able to connect Dialogflow with GPT components. I was working with Aelyse, and with the help of TA we managed to save it. For pilot study, I prepared materials that described the four personality types and wrote the pilot study methodology section for the design document. With Sebastian and Aelyse, I was involved in running the experiment, and observing the participants for extra analysis. At the end, I was working on designing and preparing the poster for the final presentation.

Sarra During the course, I followed the initial tutorials and focused on the Dialogflow part as it was one of the main parts of our project. Then with Floortje and Aelys, I started working on a initial version of our prototype which used face detection and started the conversation with dialogflow. In a second moment, I started to add also LEDs for the interaction as we got from the lecture that this type of nonverbal communication, signals that the user is being listened to which can be a factor in the engagement and overall satisfaction of the interaction of our robot. Afterwards, I focused more on the evaluation part of our experiments with Floortje, so searched for specific papers that we could use and adapted those to draft our questionnaire that allowed us to measure engagement, satisfaction, enjoyment and gather additional feedback and the observation sheet with the HRI cues. As for the paper, I drafted a first set of research questions, hypotheses and design questions. Then I started working on the interaction design section. Conducted the ethical self-check and created the consent form according to those guidelines. Then I mainly worked on adding details in the experimental design, analysis plan and pilot study on the execution of our experiments and how the observation sheet is useful and why we decided to incorporate the elements in our questionnaire for the evaluation part.

Sebastiaan For the robot design, I contributed to various key aspects. In the initial phase, I worked with Nika on implementing and structuring Dialogflow to enable natural language interaction. I also developed ideas for personalizing the robot's interactions using personality types(MBTI) and how we could make a modular implementation of this in our robot design, such that it can be used for multiple purposes. After we found a way to do this using LLMs I worked on the prompt-engineering with ChatGPT to ensure that it would give a desired

response. I also thought of methods to assess our implementations through a pilot study.

In terms of robot and code design, I programmed movements, gestures, LEDs, and postures to make the robot's interactions more expressive and engaging. I then integrated all components, including those I created and those contributed by others, into a fully functional codebase. Additionally, I authored the introduction and interaction design sections of the design document, ensuring alignment with the project's objectives.

During the testing phase, I collaborated with Nika to run the robot's code and evaluated its performance using an incongruent interaction style with various test subjects. My contributions ensured the robot was fully functional and well-prepared for testing, experimentation, and analysis.